Full Name

Section

Fracture Report Template

In all questions: You can use Excel, Matlab, Python, or anything else you are comfortable with. Your report should be approximately this long; therefore, be concise in your explanations.

1. Show your calculations for:

   a) Derive the expression for the gross stress S. Show all your work.

   b) State the expression for the fracture toughness K_{IC}.

   c) Calculate the fracture toughness K_{IC} for each specimen.

2. The data for each group is on Moodle. For each specimen type, carry out a simple statistical analysis by calculating (i) the mean value and (ii) the standard deviation for K_{IC}.

3. Explain possible reasons for why there is a scatter in the K_{IC} values for each specimen type.

4. Comment on the quality of your experimental measurements in view of the statistical data.

5. Are the mean K_{IC} values expected to be the same for each specimen type? Why or why not?

6. In this experiment, is brittle fracture a reasonable assumption? Why or why not?

References

Appendix

Include your code & extra figures